

MODULE 10 · DOMAIN 1 - AI & ML FUNDAMENTALS

Evaluating Models & Metrics

Choosing the number that tells you whether it actually works.

DOMAIN 1 and 3 EXAM WEIGHT mixed MODULE 10 of 14

Traditional ML has **one right answer** to compare against. Generative AI usually does not.

Traditional metrics, at a glance

METRIC	ANSWERS
Accuracy	Fraction of predictions correct. Misleading on imbalanced data.
Precision	Of those we flagged, how many were right?
Recall	Of those that were positive, how many did we catch?
F1	One number balancing precision and recall.
AUC-ROC	How well the model separates classes across all thresholds.
MSE / RMSE	Average error when predicting a number. RMSE is in original units.
MAE	Average absolute error. Less sensitive to outliers.
R-squared	How much variation the model explains. Closer to 1 is better.

ON THE EXAM

First question on any metric problem: is the output a category or a number? That halves the options immediately.

The four names the exam expects

METRIC	WHAT IT MEASURES, AND WHERE
ROUGE	Overlap with a reference. Built for summarisation.
BLEU	Overlap with a reference. Built for translation.
BERTScore	Semantic similarity using embeddings, not exact words.
Perplexity	How well the model predicts text. Lower is more confident.

ON THE EXAM

ROUGE for summaries, BLEU for translation. That pairing is asked directly and is worth memorising as a pair.

ROUGE or BLEU?

ROUGE

Summarisation

- Recall-oriented: did the summary capture the reference?
- Rewards including the important content
- Used for summaries and content generation
- Higher is better

BLEU

Translation

- Precision-oriented: was the output's content in the reference?
- Penalises adding things that are not there
- Used for machine translation
- Higher is better

THE TIEBREAKER

Both compare against a human-written reference, so both need one. Neither works without ground truth to compare to.

BERTScore and perplexity

- ◆ **BERTScore** compares meaning using embeddings, so different wording still scores well.
- ◆ Fixes the main weakness of ROUGE and BLEU, which punish valid paraphrasing.
- ◆ **Perplexity** measures how surprised a model is by text. Lower is better.
- ◆ Perplexity says nothing about whether output is **true** or **useful**. Only how fluent.

ON THE EXAM

If a scenario says two answers mean the same thing but score badly on word overlap, the answer is BERTScore.

Human evaluation

- ◆ People rate outputs for helpfulness, tone, safety and correctness.
- ◆ The only reliable measure for subjective qualities. Still the gold standard.
- ◆ Slow, expensive, and it does not scale to thousands of outputs.
- ◆ Bedrock model evaluation supports human review workflows directly.

ON THE EXAM

Any scenario about brand voice, tone or 'does this feel right' points at human evaluation. Automated metrics cannot judge those.

LLM-as-a-judge

- ◆ A strong model scores another model's output against criteria you define.
- ◆ Far cheaper and faster than humans, and it scales to large volumes.
- ◆ Judges qualities word-overlap metrics cannot: relevance, coherence, faithfulness.
- ◆ Inherits the judge model's own biases. Validate it against human ratings first.

ON THE EXAM

This is the answer when a scenario needs human-like judgement at a volume humans cannot handle.

Benchmarks and Bedrock evaluation

- ◆ **Public benchmarks** such as MMLU and BIG-bench compare general model capability.
- ◆ They tell you little about performance on **your** specific task.
- ◆ **Bedrock model evaluation** compares models on your own data and prompts.
- ◆ Offers automatic scoring, LLM-as-a-judge, and human review in one place.
- ◆ This is how you answer 'which model should we use' with evidence rather than opinion.

ON THE EXAM

Benchmark scores are marketing. Evaluation on your own task is evidence. The exam prefers the second.

A/B testing and business metrics

- ◆ **A/B testing** routes some traffic to each version and compares real outcomes.
- ◆ The only way to know whether a change helps **actual users**.
- ◆ **Business metrics**: conversion, resolution rate, handling time, cost per interaction.
- ◆ Also track **latency and cost per request**. Both are evaluation dimensions.
- ◆ A model can improve on ROUGE and make the business worse.

ON THE EXAM

When a question asks how to measure the success of a deployed AI feature, the answer is usually a business outcome — not a model metric.

Carry these into the exam

- ◆ Category out means classification metrics. Number out means regression metrics.
- ◆ ROUGE for summarisation. BLEU for translation. Both need a reference.
- ◆ Same meaning, different words means BERTScore.
- ◆ Subjective quality means human evaluation. At scale, LLM-as-a-judge.
- ◆ Comparing candidate models on your data means Bedrock model evaluation.
- ◆ Measuring a live feature's success usually means a business metric.

A news organisation generates short summaries of long articles. Editors have written reference summaries for two hundred articles.

The team wants an automated score to track whether summary quality improves between model versions.

Which metric fits?

- | | |
|-------------------------------------|---------------------------------------|
| ☐ A BLEU | ☐ B ROUGE |
| ☐ C F1 score | ☐ D Perplexity |

◆ PAUSE AND DECIDE

Which metric fits?

A BLEU

Built for translation. Wrong task.

C F1 score

A classification metric. Nothing is being classified.

B ROUGE

Built for summarisation, and references exist.

D Perplexity

Measures fluency, not whether content was captured.

ON THE EXAM

Two conditions are satisfied: the task is summarisation, and human references exist. ROUGE needs both.

A bank builds a model to flag money-laundering transactions. Around one in four thousand transactions is suspicious.

A compliance officer reviews every flagged case. The bank's greatest concern is a laundering transaction passing unnoticed.

Which metric should the team optimise?

- | | |
|-------------------------------------|--------------------------------------|
| ☐ A Accuracy | ☐ B Precision |
| ☐ C Recall | ☐ D R-squared |

◆ PAUSE AND DECIDE

Which metric should the team optimise?

- A

 Accuracy
Flag nothing and score 99.97%. The classic trap.
- B

 Precision
Reduces false alarms. The stated fear is missed cases.
- C

 Recall
Catches the most true cases, and reviewers absorb false alarms.
- D

 R-squared
A regression metric. The output is a category.

ON THE EXAM

The imbalance is stated as a ratio so you can compute the trap: a do-nothing model is 99.975% accurate.

A support team generates answers to customer questions. Reviewers note that many answers are correct and well written, but score poorly because they use different wording from the reference answers.

The team wants scores that reflect meaning rather than phrasing.

Which metric should they adopt?

☐ A ROUGE

☐ B BERTScore

☐ C BLEU

☐ D Accuracy

◆ PAUSE AND DECIDE

Which metric should they adopt?

- A

 ROUGE
Word overlap. This is the metric causing the problem.
- B

 BERTScore
Embedding-based, so paraphrasing scores correctly.
- C

 BLEU
Also word overlap. Same weakness as ROUGE.
- D

 Accuracy
Needs a single right answer. Open text has many.

ON THE EXAM

The scenario describes the exact weakness of overlap metrics: valid paraphrasing punished. BERTScore exists to solve it.

A company must choose between two foundation models for a legal drafting assistant. Both perform similarly on published benchmarks.

Leadership wants evidence based on the firm's own documents and prompts before committing.

What should the team do?

- ☐ A Choose the model with the higher MMLU score
- ☐ B Run Bedrock model evaluation on their own prompts
- ☐ C Choose the larger model
- ☐ D Measure perplexity on both

◆ PAUSE AND DECIDE

What should the team do?

- A

 Higher MMLU
General capability. The scenario asks about their own task.
- B

 Bedrock model evaluation
Compares candidates on the firm's own data and prompts.
- C

 The larger model
Size is not evidence, and costs more.
- D

 Perplexity
Measures fluency, not drafting quality.

ON THE EXAM

The phrase 'their own documents and prompts' rules out every general measure. Only task-specific evaluation answers it.

A retailer launches a generative support assistant. Six weeks in, ROUGE scores against reference answers have improved by twelve percent after a prompt change.

However, the proportion of chats escalating to a human has also risen.

What should the team conclude?

- ☐ A The prompt change worked; ROUGE improved
- ☐ B ROUGE is the wrong measure of success here
- ☐ C Escalation rate is unrelated to assistant quality
- ☐ D Switch to BLEU for a clearer signal

◆ PAUSE AND DECIDE

What should the team conclude?

- A

 The change worked
A model metric improved while the business outcome worsened.
- B

 ROUGE is the wrong measure
Escalation rate is the real success signal here.
- C

 Escalation is unrelated
It is the clearest signal that users are not being helped.
- D

 Switch to BLEU
Another overlap metric. Same blind spot.

ON THE EXAM

Two numbers disagreeing is the point. When a model metric and a business outcome conflict, trust the business outcome.

A media company must review the tone and brand consistency of forty thousand generated product descriptions each month.

Human reviewers can manage a few hundred. Tone cannot be captured by any word-overlap metric.

What approach fits?

- ☐ A ROUGE against reference descriptions
- ☐ B Human evaluation of all forty thousand
- ☐ C LLM-as-a-judge, validated against human ratings
- ☐ D Perplexity scoring

◆ PAUSE AND DECIDE

What approach fits?

- A

 ROUGE
The scenario states tone is not captured by overlap metrics.
- B

 Human evaluation of all
Stated as impossible. Reviewers manage hundreds, not thousands.
- C

 LLM-as-a-judge
Human-like judgement at machine scale, checked against humans.
- D

 Perplexity
Measures fluency. Says nothing about brand tone.

ON THE EXAM

The scenario rules out two options in its own wording. Read those constraints as instructions, not background.

An energy company predicts next-day electricity demand in megawatt-hours. The forecasting team wants an error measure expressed in megawatt-hours so operators can interpret it directly.

Occasional extreme demand spikes matter greatly and should be penalised heavily.

Which metric fits best?

☐ A MAE

☐ B RMSE

☐ C F1 score

☐ D R-squared

◆ PAUSE AND DECIDE

Which metric fits best?

A MAE

In the right units, but treats large errors like small ones.

C F1 score

Classification. The output is a quantity.

B RMSE

Original units, and squaring punishes big misses.

D R-squared

Explained variation, and it is unitless.

ON THE EXAM

Two requirements narrow it precisely: interpretable units rules out R-squared, and penalising spikes rules out MAE.

A travel company has two candidate prompts for its booking assistant. Offline evaluation scores them almost identically.

The product manager wants to know which one actually leads to more completed bookings.

What should they do?

- ☐ A Pick the one with the marginally better offline score
- ☐ B Run an A/B test in production
- ☐ C Measure perplexity on both prompts
- ☐ D Ask an LLM judge to pick the winner

◆ PAUSE AND DECIDE

What should they do?

- A

 Marginally better score
The scores are near identical and measure the wrong thing.
- C

 Perplexity
Fluency of text. Not booking completion.
- B

 A/B test in production
Compares real completed bookings between versions.
- D

 LLM judge
Judges output quality, not whether customers booked.

ON THE EXAM

The question asks about a real user outcome. Only a live test with real traffic can answer that.